

**Victoria Junior College**  
**Biology Department**  
**2015 Prelims Biology (H2) - Proposed answers**



A! – accept    R! – reject    (;) -1/2m

**Section A: Structured Questions**

**1 (a)** Describe the primary structure of collagen. [3]

- Each polypeptide chain is a kinked helix;
- Has repeating sequence (Gly-X-Y);
- where X is usually proline and Y often hydroxyproline;
- Every 3rd aa is glycine;
- Contains modified aa hydroxyproline and hydroxylysine;
- amino acids held by peptide bonds;

**(b)** State the organelles in which steps **(i)** and **(ii)** occur in. [2]

Step (i) – rough ER;;

Step (ii) – GA;;

**(c)** Explain how steps **(iv)** and **(v)** contribute to the high tensile strength of collagen. [2]

- staggered arrangement of tropocollagen to ensure no weak areas;;
- covalent crosslinks (between tropocollagen) to strengthen structure;;

**(d)** Suggest why the assembly of collagen takes place outside the cell. [2]

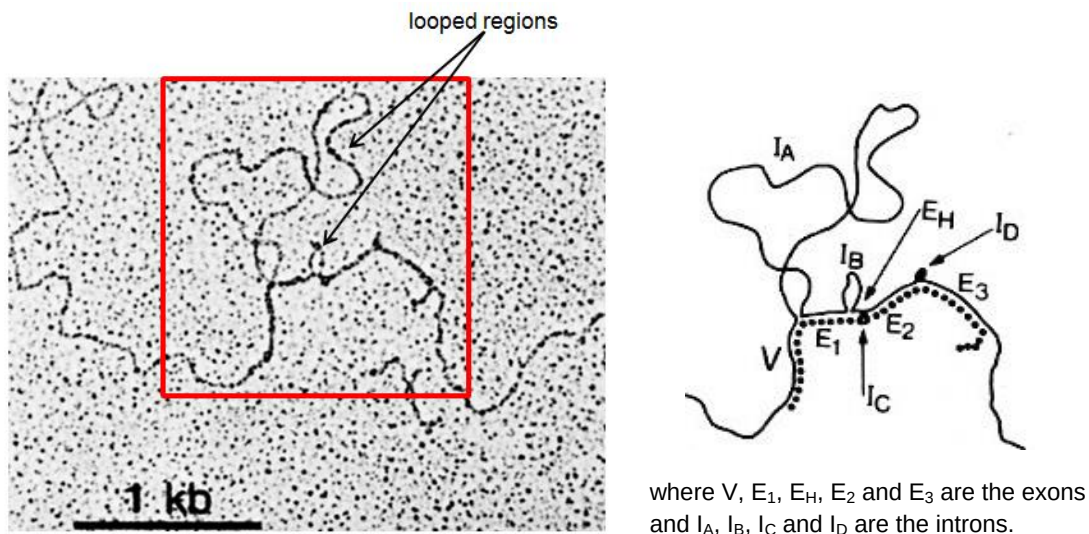
- **Cleaving of the ends of the tropocollagen** before assembly into collagen fibres is performed by **enzymes found only outside** of the cell;;
- Collagen molecule is **too large to pass through the cell surface membrane** and has to be assembled outside the cell;;
- Assembly of collagen outside the cell **allows alignment of microfibrils /** formation of bonds between the microfibrils;;

**(e)** Describe two structural differences between tropocollagen and haemoglobin. [2]  
 [1m for each comparison]

| <b>Tropocollagen</b>                                                                                  | <b>Haemoglobin</b>                                                                                       |
|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Tropocollagen is made up of 3 kinked helix                                                            | Haemoglobin is made up of 2 alpha and 2 beta globin chains/ 4 sub-units / polypeptides                   |
| Fibrous, long, strand-like structure                                                                  | Compact, highly folded, globular/ spherical structure                                                    |
| Tropocollagen has hydrophilic (R) groups on inside and hydrophobic (R) groups on outside of molecule; | Haemoglobin has hydrophobic (R) groups on inside and hydrophilic (R) groups on outside of molecule;      |
| Interchain hydrogen bonds between the polypeptide chains                                              | Hydrogen bonds, ionic bonds, hydrophobic interactions and disulfide bonds-between the polypeptide chains |

[Total:11]

- 2 (a) The image in **Fig. 2.1** is an electron micrograph where the boxed area is a mRNA-DNA hybrid molecule formed by mixing an antibody-coding messenger RNA (mRNA) from an immune cell with DNA that was made single-stranded. It consists of single and double-stranded regions. The bar represents the length of 1000 bases.



**Fig. 2.1**

(Source: <http://users.rcn.com/jkimball.ma.ultranet/BiologyPages/T/TTranscription.html>)

- (i) Account for the mRNA-DNA hybrid molecule in the boxed area. [3]
- **mRNA** has **introns excised and exon spliced together** during post-transcriptional modifications;;
  - the mature mRNA will hybridise with only the **exon** regions of the single-stranded DNA via **complementary base pairing**;;
  - **loop regions** are the unhybridized/ unpaired regions/ **introns on the DNA**;;
  - The **unhybridized portion** of the **mRNA** is its **poly(A) tail** which is added to the **3' end** of transcript by a poly (A) polymerase;;
- (ii) Different immune cells produce different types of antibodies. When an antibody-coding mRNA from another immune cell is mixed with the **same single stranded DNA**, **more looped regions** are formed.

Suggest why there is a difference in the mRNA-DNA hybrid molecules formed. [2]

- **Alternative splicing** where some **alternative exons are removed**, besides the introns;;
- Even though it is the **same gene**, **different forms of mRNA/ splice variants/ transcript variants** are produced from the **2 different immune cells**;; (A! shorter mRNA formed in this immune cell compared to the previous cell even though the same gene is transcribed)

(b) In order to study the process of translation, these mRNAs are introduced into different *E.coli* bacterial cells.

(i) State one other type of nucleic acid, besides mRNA, that must be present in the bacteria in order for translation to occur. Describe clearly its role in translation. [2]

- **tRNA;**
- **brings the correct amino acid (to the ribosome) to be incorporated into the growing polypeptide chain (by complementary base-pairing between anticodon of tRNA with codon of mRNA);;**

OR

- **rRNA;**
- **sequence on the mRNA binding site of small subunit of ribosome which complementary base-pair with the Shine-Dalgarno sequence on the 5' end of the mRNA to initiate translation;;**

(ii) What are the major differences between translation in the *E. coli* compared to that of the immune cells. [2]

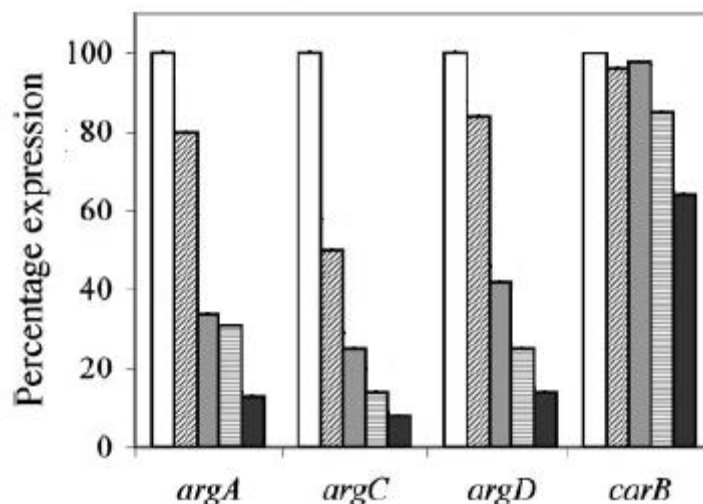
[1m for each comparison]

| Feature                                     | <i>E. coli</i>                                                                                                                                                                                                       | Immune cells                                                                                                                                                                                                                                                                                                             |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>*Coupling with transcription</b>         | Yes, <b>coupling of translation with transcription</b> due to absence of nuclear envelope.                                                                                                                           | No, <b>separate processes</b> as transcription occurs in the nucleus while translation takes place in the cytoplasm.                                                                                                                                                                                                     |
| Occurance                                   | On <b>70S</b> ribosomes                                                                                                                                                                                              | On <b>80S</b> ribosomes                                                                                                                                                                                                                                                                                                  |
| Amino acid that binds to the initiator tRNA | <b>formylmethionine</b>                                                                                                                                                                                              | <b>methionine</b>                                                                                                                                                                                                                                                                                                        |
| Initiation of translation                   | Small ribosomal subunit binds to <b>Shine-Dalgarno sequence on 5' end of mRNA;</b><br><br>→ <b>initiator tRNA binds to start codon</b><br><br>→ large ribosomal subunit binds to form translation initiation complex | Small ribosomal subunit binds to <b>initiator tRNA then to finder sequence on 5' end of mRNA;</b><br><br>→ <b>small ribosomal subunit with bound tRNA moves downstream to start codon</b><br><br>→ <b>eukaryotic translation initiation factors</b> bring large ribosomal subunit to form translation initiation complex |
| Number of polypeptide chains translated     | <b>Multiple polypeptide chains with related functions</b> can be translated from a <b>polycistronic mRNA</b>                                                                                                         | Only <b>one polypeptide chain</b> is translated from a <b>monocistronic mRNA</b>                                                                                                                                                                                                                                         |

- (iii) Explain why, despite the differences, it is still possible to synthesise the protein in the *E.coli* bacterial cell. [2]
- **Genetic code is universal** regardless of whether it is prokaryotic or eukaryotic;;
  - **same triplet of bases/ codon codes for the same amino acid;;**
  - Mature mRNA **does not have introns/** has undergone post-transcriptional modifications hence bacterial protein synthesising machinery can translate it to synthesise the protein;;

[Total: 11]

- 3 In *E. coli*, arginine is synthesized in eight enzymatic steps. **Fig. 3.1** shows the results of an experiment carried out to determine the effects of adding arginine on the expression of some of the genes involved in its synthesis. The initial gene expression was measured by determining the mRNA produced at time zero. This is taken as 100% (white bars). All the values are relative to this initial value taken every 30 seconds over the next 2 minutes.



**Fig. 3.1**

(Adapted from <http://mic.sgmjournals.org/content/journal/micro/10.1099/mic.0.29088-0?crawler=true&mimetype=application/pdf> )

- (a) Using data from *argA*, *argC* and *argD* in **Fig. 3.1** and your knowledge of how different types of operons are regulated, explain how arginine is able to control the expression of these genes. [4]
1. Gene expression decreased from 100 % at 0 min to 10 % at 2 min;;
  2. Reference to this being a repressible system;
  3. Where expression of the genes are constitutively active (100% at time 0);
  4. Addition of arginine → arginine act as a co-repressor;
  5. and binds to the repressor;
  6. This changes the conformation of the repressor to make it active;
  7. it binds to the operator;
  8. Blocks the binding of the RNA polymerase to the promoter/ prevents the RNA polymerase from accessing these genes;
  9. the structural genes are not transcribed to form mRNA → gene expression ↓;

- Award ½ mark for end product inhibition/ negative control but no full mark if repressible operon is not mentioned
- A! negative feedback but not the preferred wording;

**(b)** It has been suggested that some of the genes involved in arginine synthesis are not organized into one single operon. Instead, they are scattered over different regions of the bacterial chromosome.

**(i)** Use evidence from **Fig. 3.1** to support the statement above. [3]

- *argA*, *argC* and *argD* respond to arginine (corepressor) to the same/ comparable extent from 1 minute onwards suggesting that they are under the control of the same promoter and operator;;
- However, *carB* does not respond significantly to arginine in the same period of time → found in a different part of the bacterial chromosome, under the control of a different promoter ;;
- As gene expression decrease only to 60% at 2 min compared to about 10% for the other genes;;

A! If *argC* is placed on a different operon from *argA* and *argD*, with valid argument;

**(ii)** Suggest how expression of genes not in the same operon can be regulated by arginine. [2]

- By other intermediate products produced by enzymes in the arginine synthesis pathway;;
- These products are involved in the regulation/ inhibition of gene expression by binding to the same repressor proteins but changing them to different shapes/ conformations which bind to different operators;;

[Total: 9]

**4 (a) (i)** Describe how the DNA sequences found in region **Q** may differ from region **V**. [1]

- Q consists of tandem repeating sequences while V may consists of unique, non-repeating/ gene sequences;;
- OR Q is non-coding while V is coding sequences;;

**(ii)** Describe the packing of the DNA at region **Q** at the stage shown in **Fig. 4.1**. [3]

- Solenoid structure forms looped domains;
  - anchored by non-histone scaffolding protein to form a 300nm fiber;
  - further coiling and folding to form part of chromosome structure;
  - Ref DNA methylation, histone methylation, histone deacetylation;
  - Densely/highly coiled;
  - to form heterochromatin;
  - Ref association of DNA with kinetochore proteins;
- Max3

(iii) A mutant organism with region **Q** deleted was shown to have reduced viability (e.g. lack important functions essential for survival). With reference to the functions of region **Q**, suggest an explanation for this observation. [3]

- Ref role of centromeres serving as **microtubule-organizing center (MTOC)**;
- Site for **assembly of kinetochore proteins**;
- **Which allows for attachment of spindle fibres**;
- Deletion of centromere in chromosomes **prevent proper attachment of spindle fibres**;
- And causes **unequal separation of chromosomes / non-disjunction** into daughter cells;
- Some daughter cells would not have the full assortment of chromosomes;
- ref missing critical gene sequences coding for proteins that play essential functions for survival;;

Max 3

(b) At another region on the same chromosome, a type of specialized DNA sequences was shown to be protected from exonuclease activity by a process known as 'heterochromatization'.

(i) Explain the significance of protecting these sequences from exonuclease activity. [2]

- Sequences are **telomeres** and serves to buffer against loss of critical coding regions from end-replication problem;;
- Protection of telomeres ensures that cell can continue dividing for extended periods of time without inducing apoptosis;;

(ii) Suggest another way in which these sequences may be protected from exonuclease activity. [1]

- **Loop back** to prevent shape of telomere ends being complementary to exonuclease active site / prevent the telomeres from being recognized by exonuclease;;

[Total: 10]

5 Specific inherited mutations *BRCA1* and *BRCA2* increase the risk of female breast and ovarian cancers. Both *BRCA1* and *BRCA2* have DNA repair functions. Breast and ovarian cancers associated with *BRCA1* and *BRCA2* mutations tend to develop at younger ages than their nonhereditary counterparts.

(a) Explain the significance of mutated *BRCA1* and *BRCA2* genes in the development of cancer. [3]

- Both are **tumour suppressor genes**; (as they are involved in DNA repair)
  - **Loss of function** mutations; means that non-functional protein products are produced;
  - **DNA damage is not repaired**; and cells continue to divide and **accumulate further DNA damages**;
  - leading to uncontrolled cell division as the genome is unstable;
- A! mutations in proto-oncogenes as an example of accumulating DNA damages.

(b) Explain why individuals who inherit the *BRCA1* and *BRCA2* mutations tend to develop breast cancer **at younger ages**. [3]

- Cancer is a **multistep process**;;
- Requires the **accumulation of mutations** in at least one proto-oncogene and several tumour suppressor genes;;
- Since these individuals already harbour one mutations in a tumour suppressor gene, **less time** required for the cells with inherited gene mutations to acquire these remaining mutations to become cancerous;;

Blood samples collected from 5318 Jewish subjects identified the two most common mutations in *BRCA1* gene are either deletion or insertion of bases.

(c) Explain how two different mutations can bring about the same effect? [2]

- Deletion or addition of bases results in a **frame shift** (if addition or deletion in not in multiples of 3 bases);
- change nucleotide/base sequence/different codon sequence **after the addition or deletion of bases**; Or reading frame of the mRNA is changed after the insertion or deletion of bases (Reject: the entire base sequence / codon sequence is changed)
- leads to a different amino acid sequence in the polypeptide translated;
- Hence a non-functional protein is formed;

Accept if student explain how deletion and insertion of bases can result in the formation of a stop codon, resulting in a truncated protein.

[Total: 8]

6 In pigeons, the males are the homogametic sex (ZZ) while the female are the heterogametic sex (ZW). Two genes are found on the Z chromosome. One gene codes for the colour of the feathers, which can be either brown or red. The other gene codes for intensity of the colouration, normal or lighter shade.

A brown and normal shade male is crossed with a red and lighter shade female and the resulting offspring are all brown and normal shade.

(a) (i) With the information provided, determine which allele is dominant for the colour of the feathers and the intensity of the colouration. Explain how you derived your answer. [3]

- The alleles that code for **brown (B) and normal shade (N) are dominant** over that of red and lighter shade respectively;;
- Since all the offspring have the same phenotype means the homogametic sex/male parent is a **homozygous** for brown and normal shade;
- Heterogametic female parents will pass her only **Z chromosome with the red and lighter shade alleles to all her male offspring**;
- Since all the male offspring are **heterozygotes**;
- And **expressing the brown and normal shade phenotype** suggests that these alleles are dominant, **expressed in the presence of an alternative allele**;;

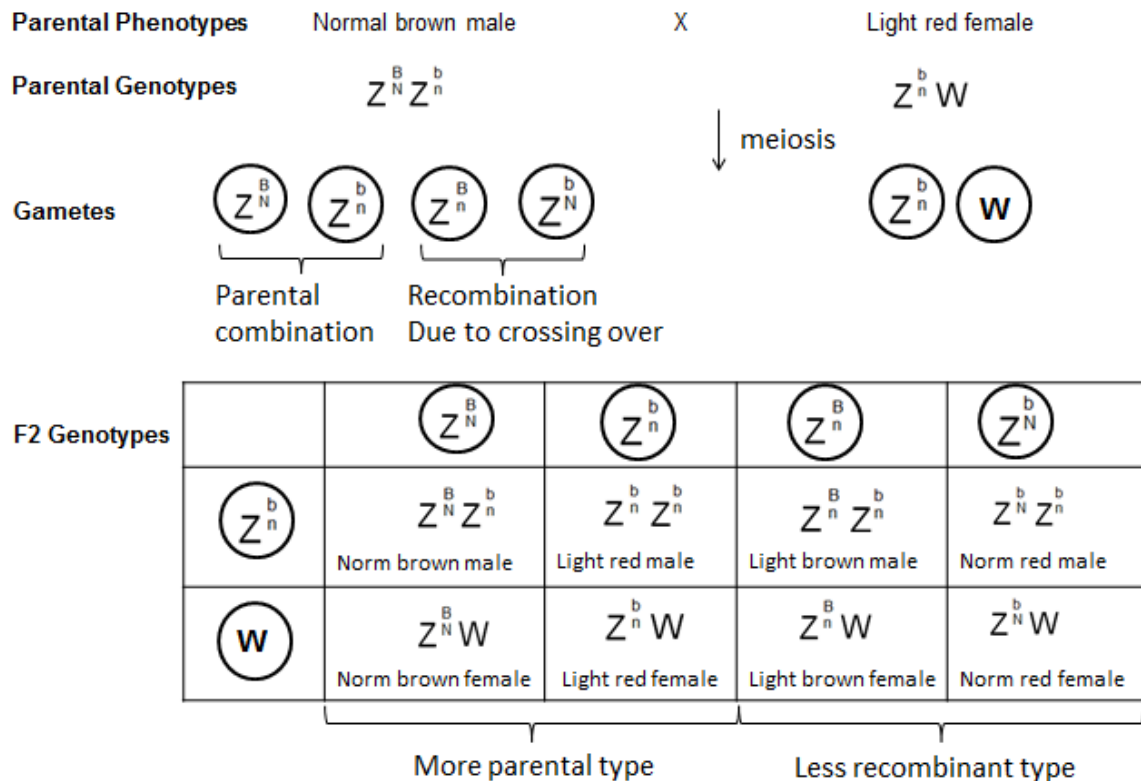
OR

- And **expressing the brown and normal shade phenotype** suggests that the **alleles for lighter shade and red must be recessive and is being masked.**;;

(ii) Using appropriate symbols for the alleles, state the genotypes of the offspring in the above cross. [1]

$$\bullet \quad Z^B_N Z^b_n \quad Z^B_N W \quad ;;$$

(b) A subsequent cross that involves a male from F1 generation and a red and lighter shade female was made. Their offspring fall into more than four phenotypic classes. Use a genetic diagram to explain the results of the cross. [5]



- Correct parental genotype and phenotype;;
- Correct gametes;;
- Correct F2 genotypes;;
- Correct corresponding F2 phenotypes;;
- Mention parental types and recombinant type in gametes and F2 genotypes;; (Mark is awarded for mentioning crossing over;)



- (c) To study the mode of inheritance of specific characteristics, controlled crosses are often carried out and the data of the offspring analysed using a chi-square test.

Explain the purpose of carrying out a chi-squared test and the conclusions that can be drawn from such a test.

- *Purpose [1]*
    - The purpose of the chi-square test is to determine if the **difference between observed and expected results was significant, or due to chance;;**
    - If the **observed data follows the expected ratio;;**
  - *Conclusions [2]*
    - If the probability of obtaining the calculated chi-square value is **greater than  $p = 0.05$ ;; OR**
    - **If the  $\chi^2_{\text{observed}}$  is less than the  $\chi^2_{\text{theoretical}}$ ;**
    - **We do not reject null hypothesis;**
    - **Difference between observed and expected data is not significant;**
    - **and is due to chance;;**
- OR
- If the probability of obtaining the calculated chi-square value is **less than  $p = 0.05$ ;; OR**
  - **If the  $\chi^2_{\text{observed}}$  is more than the  $\chi^2_{\text{theoretical}}$ ;**
  - **We reject the null hypothesis;**
  - **Difference between observed and expected data is significant;**
  - **and is not due to chance;;**

[Total: 12]

- 7 (a) (i) Identify neurones **A** and **B**. [2]

A – presynaptic neurone;

B – postsynaptic neurone;

- (ii) Support your answer to (i) with observable features shown in **Fig. 7.1**. Relate these features to their functions in the transmission of nerve impulses. [3]

- presence of synaptic vesicles;
- containing neurotransmitter;
- allow the transmission of impulse across the synapse; OR
- bind to ligand gated receptor on post synaptic membrane;
- presence of (more) mitochondria;
- to provide ATP;
- for active transport of calcium ions out of the synaptic knob/presynaptic knob; OR
- Regeneration of neurotransmitter in the presynaptic knob;

(b) With reference to **Fig. 7.2**, outline the main stages of signal transduction illustrated. [5]

- Attachment of dopamine (ligand) to the G protein linked receptor;
- Ref specific binding/ complementary shape to receptor;
- Causes a change in the conformation of the receptor;
- resulting in the activation of G protein as GTP replaces bound GDP;
- **Activate adenylyl cyclase** which converts ATP to cAMP ;
- Ref amplification of signal with production of many cAMP;
- **cAMP** phosphorylates and activates PKA;
- Opening of channel proteins (NMDA-R) via phosphorylation by PKA;
- PKA enters nucleus through nuclear pores;
- PKA phosphorylates and activates transcription factors CREB;
- which Increases/upregulates gene expression for CREM;

Max 5

[Total: 10]

8 (a) Explain the basis of tracking the lizard population to conclude if evolution has occurred. [4]

1. **Population** is the **smallest unit of evolution** ( $\frac{1}{2}$  for evolution occurs at population level)
2. The **population** is made up of **interbreeding lizards belonging to the same species/** share the same geographical area
3. **Natural selection acts on individual lizards** affecting their chances of **survival and reproductive success**
4. so that they **pass down the alleles which determine the favourable traits** to the next generation **changing the allele frequencies** of these traits **over time**
5. reference to **characteristics/** running speed and body length being **studied are heritable variations**;
6. reference to **evolution** being **measured as changes in relative proportions of heritable variations in a population in successive generations**

(b) (i) With reference to **Fig. 8.2**, describe the relationships between sprint speed and the number of offspring produced. [2]

- lizards with **greater sprint speed sire more offspring** ;
- use of **two paired figures to support relationship**, e.g. sprint speed of  $0.4525 \text{ min}^{-1}$  (A!  $0.45 - 0.455$ , m per min or per min as units), 1 offspring sired compare to  $0.6675 \text{ min}^{-1}$  (A!  $0.665 - 0.67$ ), 10 offspring sired;

(ii) Discuss whether the data shown in **Fig. 8.3** support the hypothesis that the body length is likely to reduce over time in the population. [3]

- **Yes** as there are **more offspring produced by small and intermediate length / shorter/ < 23 cm male lizards**;
- **QV**: range e.g. male lizards < 23 cm sired 1 – 10 offsprings each while those > 23 cm sired a total of only 6 offsprings;
- thus small and intermediate length male lizards will pass on their favourable

- alleles resulting in reduction of body length over time in the population;
- **No** as there is **no clear relationships**/ negative correlation **between body length and number of offspring**;;
  - **QV**: range e.g. 3 offsprings can be sired by male lizards of body lengths 20.35, 20.45, 22.05, 22.45, 22.7 and 22.95 cm;
  - thus no selective advantage of any particular body length/ the isolated case of a large male lizard siring 2 offspring may be only one large male lizard was captured, i.e. sample size too small/ AVP;

Max 2 for only one side of the argument

[Total: 9]

## Section B: Essay Questions

(;) – ½ m

### 9 (a) Outline the roles of membranes in the **regulation of blood glucose level**. [8]

#### **Detection of blood sugar level:**

- **Receptors** on cell membrane; of the
  - **β-cells** of the islets of Langerhans that detects **high blood glucose level** (higher 120 mg/ 100 cm<sup>3</sup> of blood), releasing insulin;
  - **α-cells** of the islets of Langerhans that detects **low blood glucose level** (70 mg/ 100 cm<sup>3</sup> of blood), releasing **glucagon**;

Accept if students state the normal level of blood glucose level and make reference to it in terms of high or low blood glucose levels.

#### **Synthesis of hormones insulin and glucagon and their release**

- **Rough Endoplasmic Reticulum (rER)** – single membrane organelle is responsible for the **synthesis of hormones (proteins)** / insulin and glucagon;;
- **Golgi Apparatus** – **packaging** of hormones into vesicles /formation of vesicles containing insulin or glucagon;;
- **Vesicles** containing hormones **fuse** with the membrane, thereby releasing the hormone by exocytosis, directly into the blood stream;; (mark awarded once for fusion of vesicles to cell membrane)
- **Fluid nature** of the membrane allows the fusion of vesicles; (mark awarded once only)

#### **Presence of specific receptors on cell surface membrane of effectors for the binding of specific hormones**

- Insulin attaches to the **Tyrosine Kinase Receptor (RTK)** present on **liver and muscle cells** (effectors);;
- Glucagon attaches to **G protein linked receptor** present on **liver cells**;;

#### **Specific signal transduction pathway and cellular responses that will regulate blood glucose level**

(Describe any one signal transduction pathway or cellular responses that involve membrane.)

- Example: For insulin- facilitates glucose uptake into cells
  - **Vesicles** containing **glucose transporters/ carrier proteins** transported to fusing with cell membrane, integrating the carriers into the membrane;;
  - **Increase** uptake of glucose into cell, **thereby decreasing the blood glucose level towards the norm**;;
- Example: For glucagon
  - Adenylyl cyclase present on the membrane (cytoplasmic side) that will convert ATP to cAMP which is the second messenger;;
  - Role of second messenger in phosphorylation cascade for amplification to increase concentration of enzymes to increase the concentration of glucose;;

Note: glucose must be released into the blood to raise the blood glucose concentration towards norm.

#### **Other points:**

- **Fluid nature of the membrane** allows **dimerization of the 2 subunits of the RTK**;

**(b)** State the various processes in which the concentration of active enzymes can be optimised in the cytoplasm of a eukaryotic cell. [6]

1. mRNA with a longer half-life / more stability, can be used as a template for translation longer; enable synthesis of more proteins; mRNA with longer polyA tail has longer half-life;
2. presence of 5'cap can stabilize mRNA by preventing it from being digested from 5' exonuclease; remains intact for a longer period time to be used as template;
3. masking of mRNA with inhibitor proteins (repressors) so that ribosome cannot bind and initiate translation;;
4. After enzyme has been synthesized, may not be immediately active until it has been chemically modified eg. Phosphorylation;;
5. The completed protein may need to undergo proteolytic cleavage; activation of protein after a sequence of amino acids is removed / cleaved;
6. After the enzyme has completed its function, ubiquitin can be added to the enzyme to mark it for degradation by proteasome;;

**(c)** Explain the mechanisms that genetic variation may be preserved in a natural population. [6]

1. **Diploidy** – eukaryotes are **diploid** or have **two copies of alleles present for each locus**; *(Each gene can have more than 2 alleles but at any one time, a diploid organism can only have two. So it is not correct to say that the each gene has 2 alleles.)*

- Recessive alleles can be preserved through heterozygous individuals/ recessive alleles may persist by propagation in heterozygotes in population;
- As they are **masked by the dominant alleles** of the genes;
- And hidden from selective pressure/ recessive phenotypes may be selected against;
- Only alleles that are expressed /homozygous and heterozygous dominant and homozygous recessive will be subjected to natural selection;
- recessive alleles being passed on to the offspring when heterozygotes propagate;  
(max 2)

2. **Balancing selection** occurs when natural selection maintains 2 or more forms in population; (Note: each form is a result of a particular genotype i.e combination of alleles)

- **Heterozygote advantage** where heterozygous have greater fitness/ **selective advantage over both kinds of homozygotes**;;
- Both alleles/ both dominant and recessive alleles (at that locus) will be maintained by natural selection/ natural selection favours two or more alleles;
- **Frequency-dependent selection**;
- When fitness of a phenotype declines if it becomes too common in the population;
- The different alleles can be maintained within the population; or frequency of different alleles oscillates over time

(max 3)

### 3. Neutral variation;

- When variation in DNA in population has little or no impact on reproductive success;
  - Variation occur as a result of mutation in the non-coding or coding regions of genome;
  - For coding region, mutation resulting in variation in proteins that ultimately has no effect on protein function and reproductive fitness eg. substitution - degeneracy of genetic code;
  - Neutral mutations do not affect phenotype of the organism / confer selective advantage or disadvantage to the organism, thus are not acted on by natural selection;
- (max 2)

### 10(a) Explain the mode of action of aminoacyl tRNA synthetase. [8]

[Each bullet is ½ m]

- Aminoacyl tRNA synthetase catalyses the formation of a charged tRNA.
- They are enzymes that are biological catalysts that greatly increase the rate of a chemical reaction without themselves being changed at the end of the reaction.
- Enzymes are highly specific in the reactions they catalyse. The reactant they act on is called the enzyme substrate; here they are both amino acids and tRNA.
- A specific Aminoacyl tRNA synthetase binds a particular amino acid and also a molecule of ATP at its active site.
- Effective collision between the specific substrate molecules and the enzyme causes the substrate molecules (or substrates when there are two or more reactants), to bind to the enzyme molecule at its active site to form an enzyme-substrate complex.
- A covalent bond is formed between AMP and the carboxyl group of the amino acid, and pyrophosphate is released from the enzyme.
- They do this by lowering the activation energy required for the reaction to occur.
- A specific tRNA that has an anticodon that corresponds to the amino acid then binds to the synthetase.
- The enzyme holds the different substrate molecules in an arrangement that forces them together in the correct orientation for reaction to occur.
- Once the substrate binds to the active site of an enzyme, certain bonds in the substrate molecule may be placed under physical stress which increases likelihood of reaction.
- When the R groups of the amino acid residues at the active site of an enzyme are very close to part of the substrate, they can change the charge on the substrate.
- They can also alter the distribution of the electrons within the bonds of the substrate or cause other changes that will increase the reactivity of the substrate.
- While enzyme and substrate are joined, the catalytic action of the enzyme converts the substrate to the product (or products) of the reaction.
- The RNA sequence in the anticodon region, as well as other parts of the tRNA molecule, such as the acceptor stem, is important for recognition between the tRNA and Aminoacyl tRNA synthetase.
- The bond is broken between the amino acid and AMP, and the AMP is released
- At the same time, a covalent bond is formed between the amino acid and the 3' end of the tRNA, which is now a charged tRNA.
- The charged tRNA is then released from the enzyme, and is now ready to be used in the process of translation on a ribosome.

- Under the lock and key model, the active site has a perfect fit/complementary site for a specific substrate and once the substrate binds to the enzyme, no further modification occurs.
- Under the induced fit hypothesis however, the initial shape of the active site of an enzyme might not be complementary to the shape of the substrate molecule.
- Binding of a substrate to the active site of the enzyme causes a conformational change of the enzyme, which enables the substrate to fit more snugly into the active site.
- After the reaction has finished, the active site no longer holds the molecules tightly and releases them.
- After the products move away from the enzyme, the active site returns to the initial shape.

**(b)** Compare and contrast photophosphorylation with oxidative phosphorylation. [8]

Similarities:

- Both involve ATP synthase; in synthesising ATP;
- Both require the setting up of a proton gradient/ chemiosmosis;;
- Both involved electrons moving through the electron transport chain/ electron carriers with progressively lower energy level;;

Differences:

| Features                                | Photophosphorylation                                                                                                                                                             | Oxidative phosphorylation                                                         |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Energy conversion/<br>Light requirement | Light to chemical;<br>Light required;                                                                                                                                            | Chemical to chemical;<br>No light requirement;                                    |
| Location                                | Thylakoid membrane (of chloroplast);                                                                                                                                             | Inner Mitochondrial membrane;                                                     |
| Occurs in the process                   | Light dependent reaction of photosynthesis                                                                                                                                       | Aerobic respiration                                                               |
| Sources of electrons                    | Water and reaction centre of Photosystems I/II                                                                                                                                   | Reduced NAD and reduced FAD;                                                      |
| Final electron acceptor (of ETC)        | NADP;                                                                                                                                                                            | Oxygen;                                                                           |
| Role of coenzymes                       | NADP is reduced in the light reaction as final electron acceptor;                                                                                                                | NAD is regenerated for use in glycolysis, link reaction and Krebs's cycle;        |
| Roles of O <sub>2</sub>                 | By-product of photolysis;                                                                                                                                                        | Final electron acceptor;                                                          |
| Role of water                           | Photolysis of water provides:<br>(i) H <sup>+</sup> which will then accumulate in the thylakoid space and electrons to replace the electron lost at the primary reaction centre; | By-product of reduction of O <sub>2</sub> (oxygen accepts e and H <sup>+</sup> ); |
| Proton reservoirs                       | Thylakoid space;                                                                                                                                                                 | Intermembrane space;                                                              |

|                                            |                                                                                                               |                                                                                                            |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>Direction of H<sup>+</sup> movement</b> | H <sup>+</sup> is pumped from stroma to thylakoid space / lumen, moves back into stroma through ATP synthase; | H <sup>+</sup> is pumped from matrix to intermembranal space, moves back into matrix through ATP synthase; |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|

(c) When the human genome was sequenced, it was found to contain a surprising high proportion of viral sequences. Suggest how this could have happened. [4]

1. **retrovirus inserting its genome;;**
2. in **human populations** through **past infections;;**
3. viral sequences are found in the **non-coding region;;**
4. Viral infection **does not lead to death** of the individual;;
5. Molecular differences are **selectively neutral/** not under selection pressure;;
6. and **do not affect fitness** of the individual/ idea that the person is able to survive and reproduce;;
7. can be **retained in the human genome over time;;**